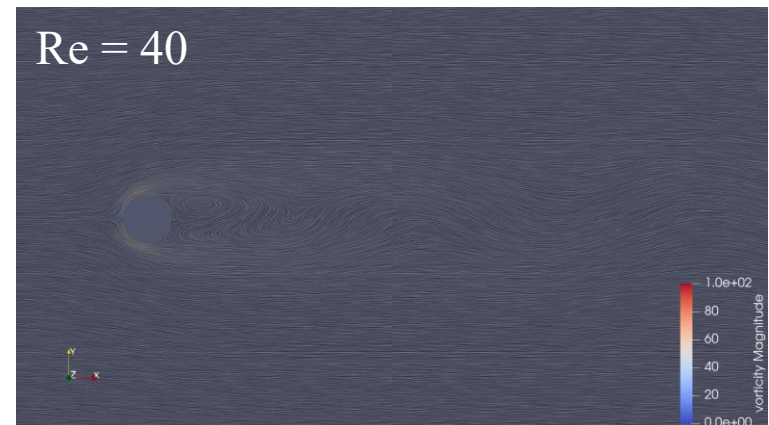
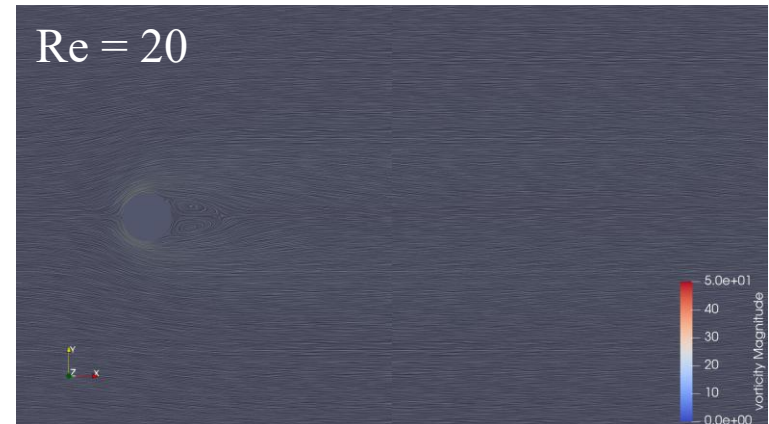
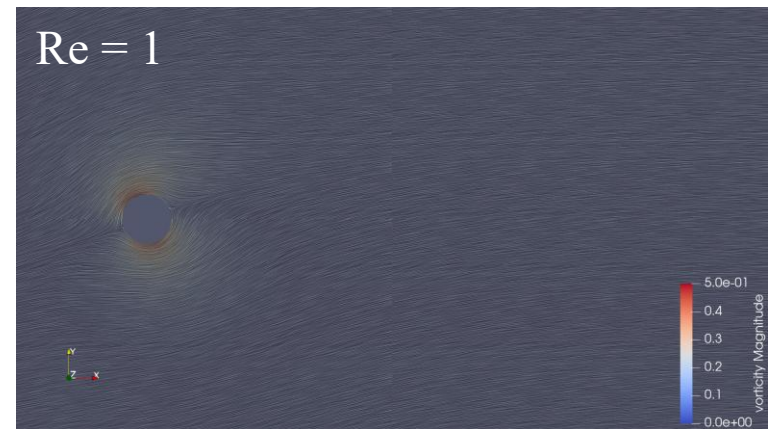
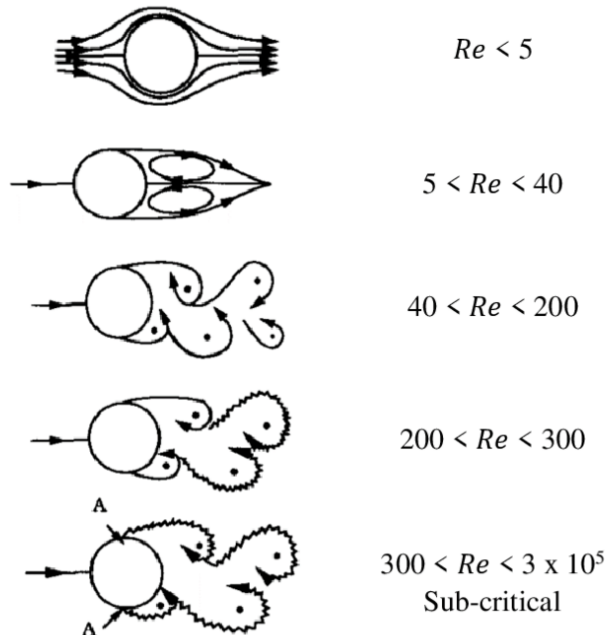


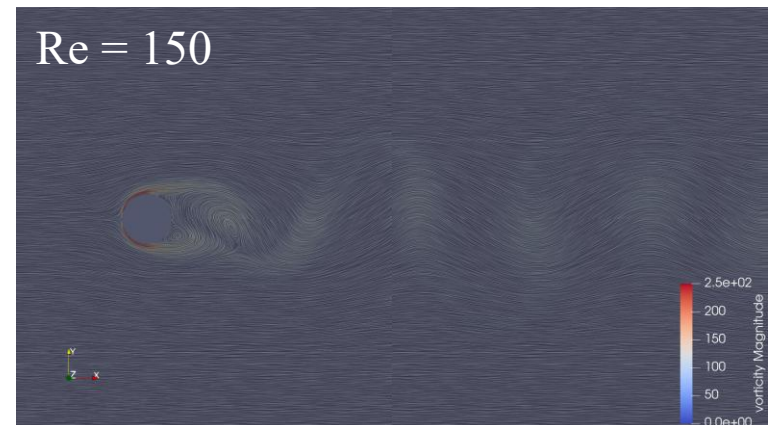
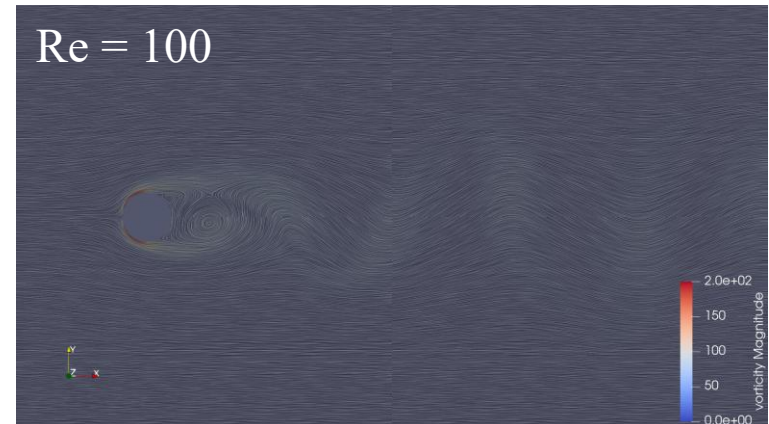
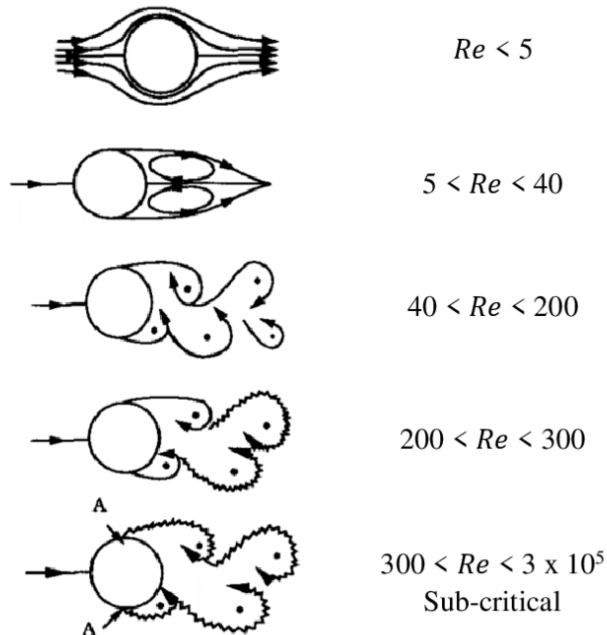
# Simulation Results

- **Solver:** PISO (Laminar)
- **Schemes:** Backward (time), LUST (convection)
- **Reynolds number:** 1, 20, 40, 50, 100, 150
- **Run controls:**
  - Simulation time = 10 s ,  $\Delta t = 1\text{e-}3$  s,
  - Output interval = 0.025 s (25 steps)
  - Used = 8 s data for shedding,
  - 3 s ~ 8 s data for post-wake data
- **Verified reference:**
  - Duranay et al. (2022),*
  - An Experimental Study on Vortex-Induced Vibrations at TRSL2 and TRSL3 Flow Regimes



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# Post-Wake Macro Output Guide

- Automated post-processing of cylinder wake simulations — extracts force data, performs FFT of  $C_L$ , computes Strouhal number, and aggregates results for multiple Reynolds cases.

## 1. Per-case output (in each case or in central folder)

---

| File / Figure           | Description                                                                                 |
|-------------------------|---------------------------------------------------------------------------------------------|
| <i>forces_clean.csv</i> | Time history of ( $C_d$ , $C_L$ , $C_m$ ) after transient trimming                          |
| <i>summary.json</i>     | Case summary: Re, U, $C_{d_{mean}}$ , $f_{peak}$ , St, % errors vs reference                |
| <i>CL_time.png</i>      | Plot of $C_L(t)$ vs time, showing periodic vortex shedding                                  |
| <i>CL_psd.png</i>       | Power spectral density of $C_L$ ; vertical dashed line shows dominant frequency used for St |

---

## 2. Central aggregate folder (e.g., *post\_wake\_all*)

---

| File                     | Description                                                             |
|--------------------------|-------------------------------------------------------------------------|
| <i>aggregate.csv</i>     | Tabular summary for all scanned cases (Re, St, $C_{d_{mean}}$ , errors) |
| <i>aggregate.json</i>    | Same as CSV in JSON format                                              |
| <i>Re_St_scatter.png</i> | Reynolds number vs Strouhal number (with reference curve overlay)       |
| <i>Re_Cd_scatter.png</i> | Reynolds number vs mean drag coefficient (with reference data overlay)  |

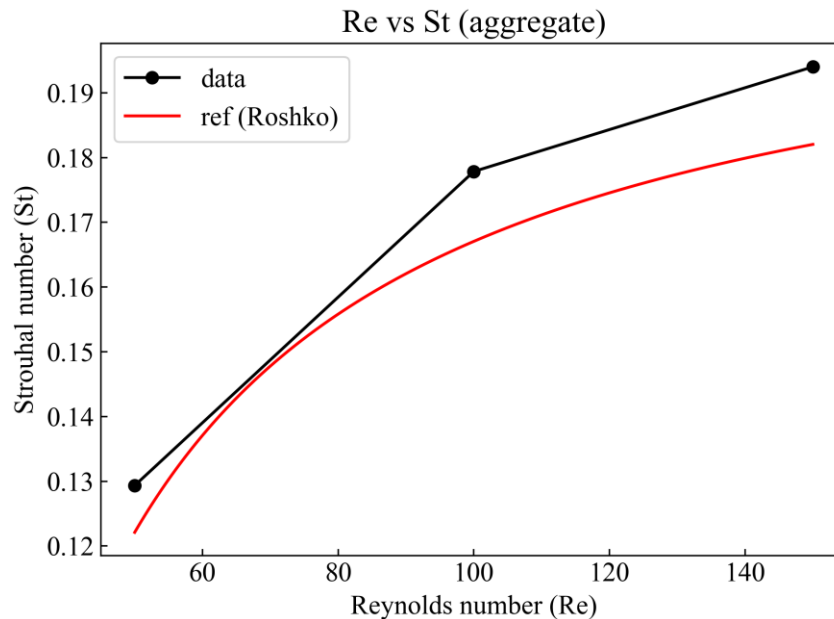
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# Re vs St (Strouhal Number)

- Shows how the vortex shedding frequency changes with Reynolds number.

$$St = \frac{fD}{U_\infty}$$

- It is the primary non-dimensional indicator of vortex shedding periodicity.
- For  $Re < 47$ ,  $St \sim 0$  (steady wake). As  $Re$  increases,  $St$  rises and tends to  $\sim 0.2$  in the laminar vortex shedding regime.

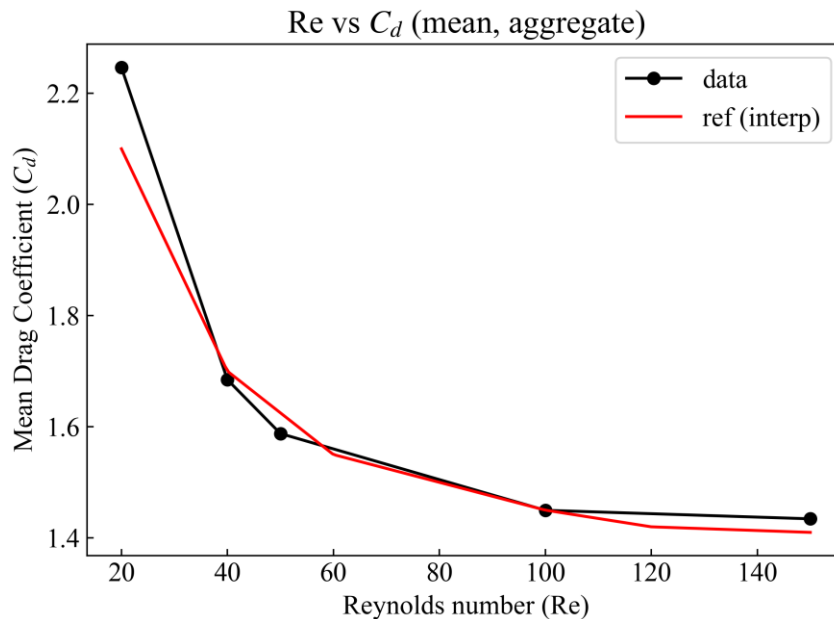


# Re vs $C_d$ (Mean Drag Coefficient)

- Shows how the mean drag on the cylinder changes with Reynolds number.

$$C_d = \frac{F_d}{\frac{1}{2} \rho U_\infty^2 D}$$

- It is the primary non-dimensional indicator of vortex shedding periodicity.
- For  $Re < 47$ ,  $St \sim 0$  (steady wake). As  $Re$  increases,  $St$  rises and tends to  $\sim 0.2$  in the laminar vortex shedding regime.



# Vorticity / Velocity ( $U$ ) Results Data Guide

- Stores point coordinates and vorticity components at each time step for post-processing or validation.
- File Format:
  - Header row describes each column
  - Each subsequent row contains data for one point at a given time step

|   | A        | B        | C        | D        | E           | F           | G           |
|---|----------|----------|----------|----------|-------------|-------------|-------------|
| 1 | TimeStep | Points:0 | Points:1 | Points:2 | vorticity:0 | vorticity:1 | vorticity:2 |
| 2 | 0        | -0.05    | -0.05    | 0        | -2.04E-20   | -1.83E-18   | -0.12393    |
| 3 | 0        | -0.04869 | -0.05    | 0        | 4.8184e-21  | 3.5278e-18  | 0.10665     |
| 4 | 0        | -0.04869 | -0.04902 | 0        | 5.28e-20    | 9.0578e-19  | 0.10335     |
| 5 | 0        | -0.05    | -0.04902 | 0        | -1.38E-20   | 8.8531e-19  | -0.0952     |
| 6 | 0        | -0.04869 | -0.04802 | 0        | -2.05E-20   | -3.51E-18   | 0.15708     |
| 7 | 0        | -0.05    | -0.04802 | 0        | 1.7327e-20  | 9.9185e-19  | -0.07842    |
| 8 | 0        | -0.04868 | -0.04701 | 0        | -4.43E-20   | 2.8778e-18  | 0.1888      |

| Column                                       | Description                                                                  |
|----------------------------------------------|------------------------------------------------------------------------------|
| <i>TimeStep</i>                              | Simulation time                                                              |
| <i>Points:0, Points:1, Points:2</i>          | X, Y, Z coordinates of the sample point                                      |
| <i>vorticity:0, vorticity:1, vorticity:2</i> | Vorticity vector components ( $\omega_x, \omega_y, \omega_z$ ) at that point |
| <i>U:0, U:1, U:2</i>                         | Velocity components $U_x, U_y, U_z$ at that point                            |